

NARIN SRIPHO

Automation & IoT Engineer · PLC, Control Systems & Smart Factory Integration

Licensed Associate Engineer (Electrical) — **License No. ภพก. 68797**, Council of Engineers Thailand
Certified Safety Officer, Management Level (จป.บริหาร) — Thai Ministry of Labour

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Mueang Samut Sakhon 74000, Thailand | Thai (native) & English (professional)

PROFESSIONAL PROFILE

I read and troubleshoot PLC logic on live production equipment, and build the full data path from PLC and sensors to monitoring dashboards, SCADA-style reporting and databases — so a food-processing plant can see Run/Stop, Downtime, Alarm and Performance in real time instead of finding out from paper. Twenty years on the shop floor and in the control cabinet, most recently owning PLC, IIoT and machine-monitoring delivery for a live production line end to end.

S7-1200 / TIA Portal PLC & control systems	IIoT → SCADA / DB PLC–sensor–dashboard data path	100 / min Automated sorting line	20+ yrs Engineering & automation
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KEY ACHIEVEMENTS

quantified impact

10 hrs PER WEEK RECOVERED	Built the plant's IIoT data layer end to end (n8n, Node-RED, Modbus RS485), wiring PLC and shop-floor sensors onto live dashboards and retiring paper records entirely.
400% SORTING THROUGHPUT	Programmed and commissioned a Siemens S7-1200 sorting line at 100 items/min with dual independent checkweighers, validated in 3D simulation before fabrication.
80% CAPEX REDUCTION	Designed and deployed an in-house machine vision counting system (YOLOv8 on Raspberry Pi 5 with Coral Edge TPU), eliminating 1 hour/day of manual tallying.
฿2M PER YEAR · 33% ENERGY CUT	Directed the plant's utility backbone — 404 kW solar, 1,500 kVA of transformers — clearing every Thai regulatory inspection with zero compliance delays.
฿20M+ CAPITAL PROGRAMME · 12-PERSON TEAM	Consolidated engineering, automation and IT into a single function — removing the vendor handoffs that delayed every plant-floor control or data change.

SELECTED ENGINEERING PROJECTS

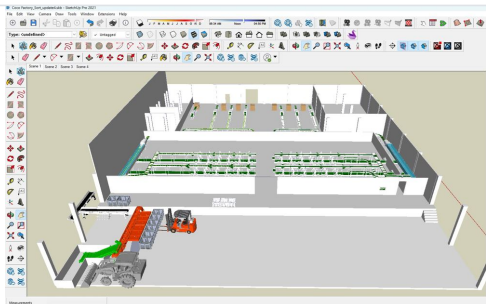
Designed and delivered in-house

01

High-Speed Sorting System

MULTI-STAGE CONVEYOR LINE · DUAL CHECKWEIGHERS · SIEMENS S7-1200

100 items/min



3D plant layout — the conveyor runs and packing lines the sorting system was designed into.

PROBLEM	APPROACH	RESULT
Sorting throughput capped the line, and a mechanical redesign at this scale normally means discovering clearance and load faults on the shop floor — after fabrication.	Specified a multi-stage conveyor with dual independent checkweighers, driven by Siemens S7-1200 PLCs and pneumatic actuators — validated in 3D structural and kinematic simulation before cutting steel.	Throughput raised 400% to 100 items/min. Clearance and load issues caught at design stage, avoiding an estimated ₦1,000,000 in rework.

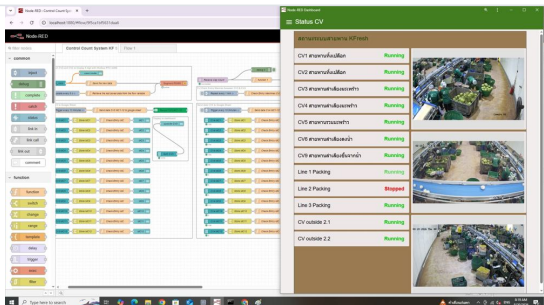
SolidWorks · Kinematic simulation · Siemens S7-1200 / TIA Portal · Pneumatic actuators · Checkweigher integration

02

Plant IIoT Data Layer

N8N · NODE-RED · MODBUS RS485 · SELF-HOSTED VPS

10 hrs/week



Live Node-RED dashboard — per-machine counts pulled over Modbus RS485, every line shown running or stopped in one glance.

PROBLEM	APPROACH	RESULT
Shop-floor readings lived on paper and in spreadsheets — visible only after the fact, and only to whoever typed them up.	Wired PLC and IoT sensors into centralised databases over Modbus RS485, orchestrated with n8n pipelines on a Docker/Portainer VPS, and published live Node-RED dashboards for operators.	Paper records retired and 10 hours/week of administrative work recovered — the plant reads its own state live instead of reconstructing it after the fact.

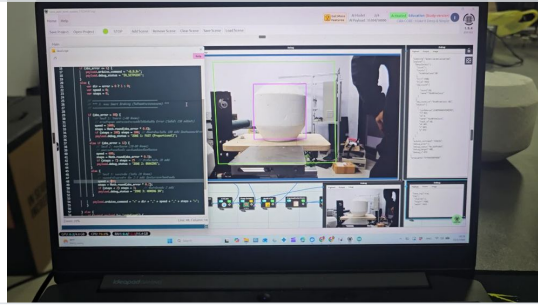
n8n · Node-RED · Modbus RS485 · Docker / Portainer · Webhooks & REST · Google Apps Script

03

Edge AI Vision Counter

YOLOV8 · CORAL EDGE TPU · RASPBERRY PI 5 · CLOUD LOGGING

80% lower capex



Live detection on the actual line — CiRA-CORE tracking each piece under the camera, feeding the counting logic in real time.

PROBLEM

Product counting was manual — a full hour a day of tallying, with transcription error baked in. Commercial vision systems quoted at several times the budget.

APPROACH

Trained a YOLOv8 detection model via CiRA-CORE and Google Colab, deployed on-line on Raspberry Pi 5 with a Coral Edge TPU, with results auto-logged to the cloud and surfaced on operator dashboards.

RESULT

Capex cut roughly 80% versus vendor systems, 1 hour/day of manual tracking eliminated, counting logic kept in-house to retrain as products change.

YOLOv8 · Google Coral Edge TPU · Raspberry Pi 5 · CiRA-CORE · Google Colab

04

Non-Blocking Control Architecture

ESP32 · FREERTOS · STATE-MACHINE DESIGN · FOOD PROCESSING

Real-time safe

PROBLEM

Cloud logging on a food-processing controller competes with the control loop — a blocking transmit means a missed real-time deadline on live equipment.

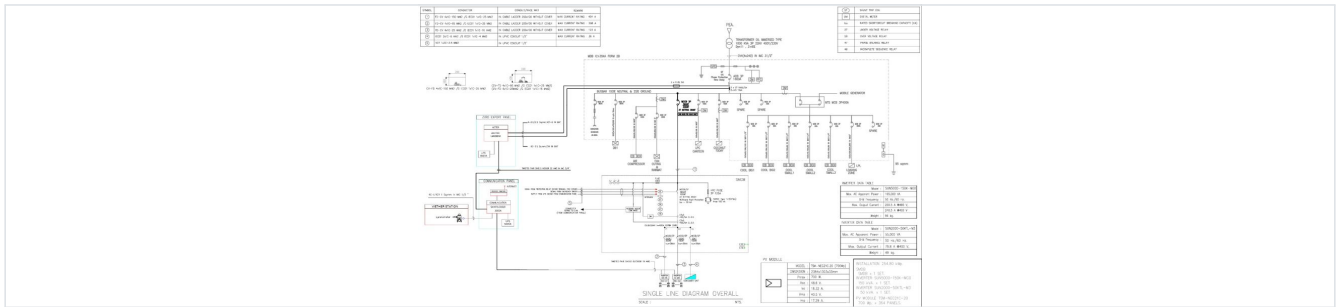
APPROACH

Built the controller on ESP32 + FreeRTOS with data transmission isolated as a separate background task, applying state-machine design, ISRs and hardware debouncing; HMI integrated with relay boards over Modbus RS485 on an isolated control LAN.

RESULT

Cloud logging never interrupts real-time loops — telemetry without trading away deterministic control.

ESP32 / ESP8266 · FreeRTOS · ISR & debouncing · State-machine design · Isolated control network



Single line diagram — 22 kV intake to distribution, including the zero-export solar panel design.

PROBLEM

The plant carried its full energy load from the grid and had no single owner for statutory utility infrastructure — every inspection cycle risked schedule slip.

APPROACH

Took delivery end-to-end: specification, contractor supervision, regulatory inspection and commissioning across 404 kW of solar generation, 1,500 kVA of transformer capacity, HVAC, fire protection, wastewater treatment and weighbridge.

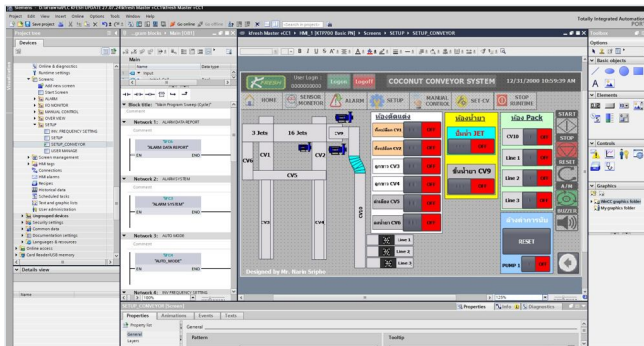
RESULT

Plant energy cost down approximately ฿2,000,000/year (33%). Every Thai regulatory inspection cleared with zero compliance delays.

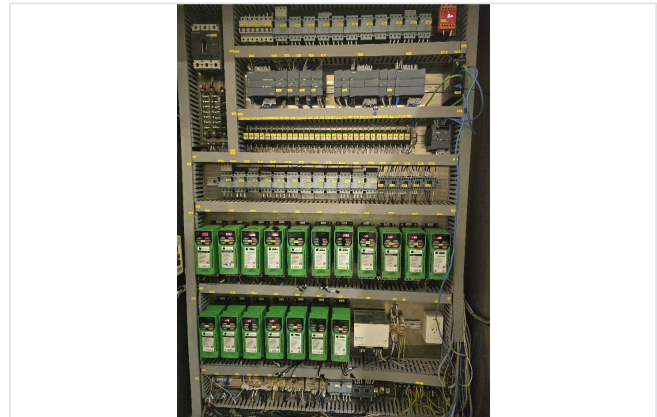
Solar PV · MV/LV distribution · HVAC · Fire protection · Wastewater treatment

SELECTED VISUALS

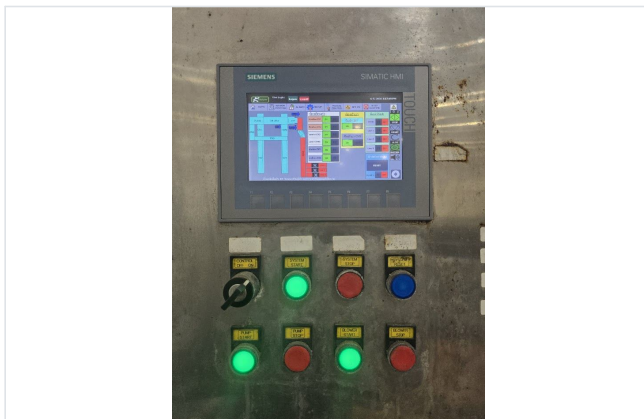
PLC programming, control cabinets and site documentation



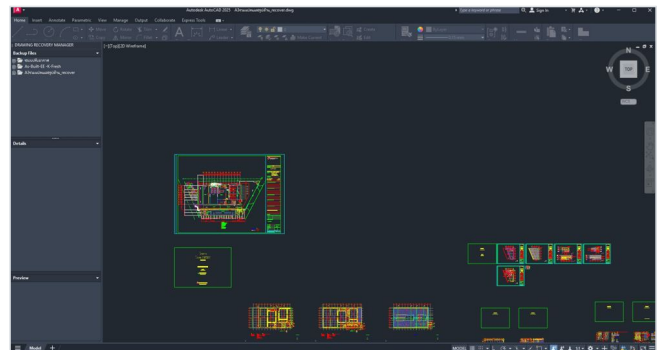
TIA Portal, mid-edit — program blocks, ladder logic and the HMI screen design for the coconut conveyor line.



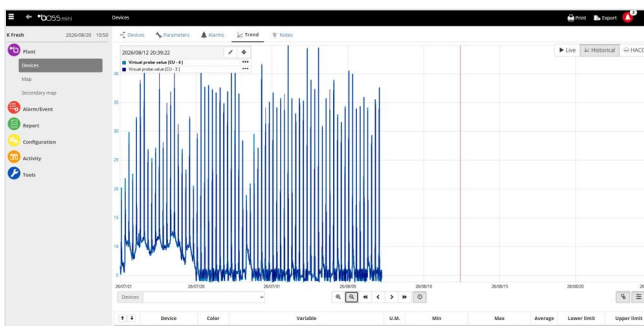
PLC control cabinet I designed and wired — Siemens S7-1200 modules, breakers, I/O relay terminals and VFDs.



Plant-floor control station — hardwired start/stop controls paired with the SIMATIC HMI.



As-built electrical & HVAC documentation — drawing sets kept current, not left to drift.



Cold-chain monitoring dashboard — continuous temperature logging with alarm thresholds for HACCP compliance.

CORE COMPETENCIES

Control cabinet to cloud dashboard

INDUSTRIAL AUTOMATION & CONTROLS

- Siemens S7-1200 (TIA Portal)
- PUT/GET remote communication
- HMI-relay integration via Modbus RS485
- Pneumatic actuator & conveyor control
- Isolated control-network architecture

IIOT, DATA & MONITORING

- n8n pipelines on VPS (Docker / Portainer)
- Node-RED dashboards (CSS/Angular)
- SCADA-style monitoring, Downtime & OEE tracking
- Webhooks & REST
- Google Apps Script

MACHINE VISION & EDGE AI

- YOLOv8 object detection
- Google Coral Edge TPU
- Raspberry Pi 5
- Training via CiRA-CORE / Google Colab

EMBEDDED & REAL-TIME SYSTEMS

- ESP32 / ESP8266
- FreeRTOS multitasking
- Non-blocking control loops
- Interrupt service routines
- State-machine design

PLANT & HEAVY ENGINEERING

- MEP infrastructure delivery
- Solar PV · Transformers & distribution
- HVAC · Fire protection · Wastewater treatment
- Contractor supervision & commissioning

MECHANICAL DESIGN & SIMULATION

- SolidWorks · SketchUp Pro
- 3D kinematic simulation
- Mechanical load & clearance validation

IT & NETWORK INFRASTRUCTURE

- MikroTik routing
- Windows SMB hardening
- CCTV & access control
- GPU deployment (RTX 5060 Ti)

SAFETY, COMPLIANCE & COORDINATION

- Safety Officer, Management Level (จป.บริหาร)
- Statutory workplace-safety accountability
- Cross-functional coordination with Production, Maintenance, IE, IT & Vendors

PROFESSIONAL EXPERIENCE

Hands-on with PLC, IIoT and machine data

IT & Innovation Manager

SEPT 2020 – PRESENT

K-Fresh Co., Ltd. — Samut Sakhon, Thailand (food-processing plant)

AUTOMATION, CONTROLS & IIOT

- Design and build mechanical conveyor systems integrated with **Siemens S7-1200 PLCs** and pneumatic actuators, taking concepts from 3D simulation through fabrication to production handover.
- Read, troubleshoot and edit PLC programs on live production equipment; diagnose Sequence Control, Interlock, Alarm, Safety and Machine Signal logic to reduce downtime.
- Integrate HMI panels with relay boards over Modbus RS485 on isolated control LANs, and produce Wiring Diagrams, Electrical Drawings and Control Circuit documentation.
- Modernise plant workflows by wiring PLC, sensor and machine data into centralised monitoring, dashboards and databases via n8n and Node-RED, replacing paper and spreadsheet reporting with live Downtime, Alarm and Performance tracking.
- Deploy machine vision and edge AI for real-time product counting — training **YOLOv8** models and running them on Raspberry Pi 5 with Coral Edge TPU — with automated cloud logging and operator dashboards.
- Develop embedded controllers on ESP32 + FreeRTOS applying state-machine design, interrupt routines and hardware debouncing for production equipment.

PLANT INFRASTRUCTURE & COORDINATION

- Supervise end-to-end delivery of major plant infrastructure — solar generation, transformers, HVAC, weighbridge, fire protection and wastewater treatment — coordinating contractors, regulatory inspection and commissioning through to handover.
- Maintain and optimise heavy production machinery with uptime accountability, and carry statutory workplace-safety responsibility as certified Safety Officer at Management Level.
- Coordinate across Production, Maintenance, IE, IT and Vendors to deliver Automation, IoT and Smart Factory initiatives that meet food-manufacturing safety and quality requirements.
- Own corporate IT and security: CCTV, access control, MikroTik routing, Windows SMB hardening, Docker/Portainer on self-hosted VPS, and GPU workstations for model training.

Manager

MAR 2009 – AUG 2020

GTA Visa Expert Co., Ltd.

- Led general administration and internal organisational operations across an 11-year tenure — accountable for process design, staffing and day-to-day management, building the operational foundation later applied to running a plant automation function.

Engineering — New Model

2005 – 2007

Supawut Co., Ltd. (บริษัท สุภาวุฒิ จำกัด)

- Designed injection and blow moulding tooling for new-model parts, owning the design from concept through to a production-ready mould.
- Set and optimised process parameters on injection and blow moulding machines; authored Work Instructions (WI) defining process settings and quality checks before handover to Production.

EDUCATION & CREDENTIALS

Licensed and certified

DEGREE

B.Eng. Electrical Engineering (Telecommunications)

Rajamangala University of
Technology Thanyaburi (RMUTT)
Graduated 2005 · GPA 3.27

PROFESSIONAL LICENSE

Associate Engineer License, Electrical

Council of Engineers Thailand ·
status active
License No. **กฟก. 68797**

STATUTORY CERTIFICATION

Safety Officer, Management Level (จป.บริหาร)

Certified under Thai Ministry of Labour
occupational safety regulations

AVAILABILITY

Let's talk about your control room

Available for **Automation Engineer**, **IoT Engineer**, **Controls Engineer** and **Smart Factory / Industrial Digitalisation** roles.

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